A Lightweight Multiscale and Multiattention Hyperspectral Image Classification Network Based on Multistage Search

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***Abstract*— Hyperspectral image (HSI) classification has become a core task in hyperspectral remote sensing interpretation, with deep learning dominating due to its ability to learn hierarchical features without manual engineering. As the model complexity has grown, manual design limitations have prompted a shift to automated approaches such as differentiable architecture search (DARTS), where the architectures are optimized for greater accuracy and efficiency. However, applying gradient-based neural architecture search (NAS) methods directly to hyperspectral classification presents several challenges. Regarding search space design, there is a lack of lightweight operators that can mitigate the spectral variability, spatial heterogeneity, and scale differences inherent in hyperspectral imagery. In terms of search strategy, the traditional DARTS approach directly derives the topology from operation weights, which can lead to suboptimal topological structures, and thus affects the performance of the network in HSI classification. In this article, to address these issues, we propose L3M, which is a lightweight multiscale and multiattention HSI classification network based on multistage search. The proposed approach introduces a novel lightweight operator to address the spectral variability, spatial heterogeneity, and scale differences in HSIs. The operation search and topology search are also decomposed into a multistage process to prevent a suboptimal network by searching for and determining the topological order of the candidate operations in a predefined operation space. L3M was validated on four public datasets, where the proposed model demonstrated a superior classification performance, compared to other lightweight models, while maintaining a low parameter count, low model complexity, and high inference speed.**

***Index Terms*— Hyperspectral image (HIS) classification, lightweight network, multiattention, multiscale stacking convolution, network topology search, neural architecture search (NAS).**

Received 23 February 2025; accepted 17 March 2025. Date of publication 20 March 2025; date of current version 4 April 2025. This work was supported in part by the National Key Research and Development Program of China under Grant 2022YFB3903404, in part by the National Natural Science Foundation of China under Grant 42301418, in part by China National Postdoctoral Program for Innovative Talents under Grant BX20230275, and in part by Hubei Provincial Science and Technology Innovation Talent Program under Grant 2024DJA020. *(Corresponding author: Yuting Wan.)*

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Digital Object Identifier 10.1109/TGRS.2025.3553147

I. Introduction

HYPERSPECTRAL remote sensing is an Earth observation technology that is capable of acquiring continuous spectral curves and integrated spectral images of surface features. It possesses the ability to finely identify the composition of surface materials. With the rapid development of hyperspectral remote sensing technology and image fusion technology, hyperspectral images (HSIs) are gradually transforming from high spectral resolution images to remote sensing images with both a nanometer-level spectral resolution and a meter-level spatial resolution [1]. These HSIs, with their rich spatial and spectral information, make various remote sensing interpretation tasks such as fine object classification possible, and are widely used in the study of urban areas, agriculture, minerals, oceans, atmosphere, deep space, and other fields [2], [3].

To address the challenges posed by the large data volume and rich band characteristics of HSIs, the traditional methods utilize the rich spectral information in HSIs and classify them based on different spectral similarity metrics between image pixels [4], [5], [6]. However, since using only the spectral information in HSIs ignores the spatial a priori knowledge and exacerbates the similarity of spectral information under feature mixing, the classification results can be characterized by serious classification failures and large-scale classification fragmentation [7]. To address these issues, HSI classification by the fusion of spatial information has been proposed [8]. Such methods rely on a combination of technical expertise and domain knowledge to create diverse human-engineered features. These features encompass aspects such as shape, texture, color, spectral characteristics, and spatial information. Notably, this integrated approach, which combines robust finite element analysis with discriminative classification [9], has demonstrated enhanced mapping accuracy when compared to using the pixel/spectral-based methods in isolation.

Handcrafted features can efficiently define diverse image attributes, but ideal feature patterns vary significantly, based on the data type [10]. Human expertise also significantly impacts their design and effectiveness during classification. To mitigate the limitations associated with handcrafted features, Salakhutdinov and Hinton [11] introduced a deep feature learning approach. Deep learning methodologies

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are utilized to hierarchically extract features from the data, constructing models with progressively more complex semantic layers. This process is completed once the model attains an appropriate data representation.

The early deep learning methodologies, akin to the traditional approaches, predominantly emphasized spectral information. Notable examples include deep belief networks (DBNs) [12], recurrent neural networks (RNNs) [13], and 1-D convolutional neural networks (1D-CNNs) [14]. However, these techniques often suffer from the "island effect" in classification outcomes. To address this limitation and leverage contextual spatial information alongside texture features, numerous spectral–spatial classification methods have been developed. For example, Sidike et al. [15] developed a novel neural network named the progressively expanded neural network that can effectively interpret hyperspectral pixels in nonlinear feature spaces with spectral–spatial information. Prominent among these approaches are 2-D CNNs (2D-CNNs) [16], 3-D CNNs (3D-CNNs) [17], [18], spatial–spectral residual networks (SSRNs) [19], fully convolutional networks (FCNs) [20], [21], [22], and spectral–spatial unified networks (SSUNs) [23], [24]. These advanced methodologies facilitate comprehensive multidimensional information extraction for hyperspectral imaging, significantly enhancing the classification accuracy, compared to the earlier methods.

Lightweight HSI classification networks have garnered significant attention due to their ability to balance a high classification accuracy with reduced computational complexity. The traditional CNN-based HSI classification models, while achieving satisfactory accuracies, often suffer from a large number of parameters and high computational demands, necessitating extensive training samples. This challenge is exacerbated by the high cost of manually labeling samples. To address these issues, researchers have developed various lightweight models. For example, Roy et al. [25] enhanced HybridSN by incorporating depthwise separable convolution (DSC) and fast convolutional blocks, reducing the model complexity and improving the classification accuracy with fewer training samples; A novel collaborative learning framework was proposed by Fang et al. [26] for semi-supervised HSI classification, integrating a lightweight 3-D convolutional network with deep clustering techniques; Cui et al. [27] made notable contributions by using 3-D depthwise convolution (DW-Conv) and focal loss to enhance the feature representation and improve the model's attention on difficult samples while reducing the computational cost; and Lv et al. [28] developed ESSINet, which is a lightweight network architecture based on a dual-pooling layer and a novel involution-2-D operator, which effectively extracts spatial–spectral features with lower computational consumption. Liu et al. [29] proposed a deep contrastive learning network (DCLN) to address the small-sample classification problem in HSI data, achieving superior performance on four public datasets. The development of these methods has been crucial for the deployment of hyperspectral algorithms on platforms with limited storage and computing resources, ensuring real-time processing capabilities for intelligent remote sensing analysis. However, the application of lightweight architectures also results in a compromise in classification accuracy.

Attention mechanisms have been introduced to HSI classification to address the challenge of combining multiscale spatial and spectral information, which often results in a poor performance for varying scales and textures. For instance, Roy et al. [30] developed an attention-based adaptive spectral–spatial kernel improved residual network (A^2S^2K-ResNet) to capture discriminative spectral–spatial features; A novel local transformer with a spatial partition restore network (SPRLT-Net) was introduced by Xue et al. [31], which captures global context dependencies and adapts to the inherent changes in HSI spatial pixels; Combining GhostNet and attention mechanisms, Wang et al. [32] proposed a lightweight neural architecture search (NAS)-guided spectral–spatial fusion neural network, which uses a multiscale ghost CNN grouped with efficient channel attention (ECA) modules to adaptively adjust the channel weights; Sun et al. [33] constructed a parallel spectral and spatial attention network (SSAN) to separately extract and fuse spectral and spatial features; Haut et al. [34] combined a residual network with visual attention techniques to identify key features; Hang et al. [35] proposed an attention-aided CNN method to adaptively weight spectral and spatial features; Hu et al. [22] first proposed scale attention and used multiple attention mechanisms to adaptively weight each channel, each pixel, and each scale perception of the feature map; Xue et al. [36] developed an attention-based second-order pooling network (A-SPN) that simplifies hyperparameter tuning and enhances explainability; and Fu et al. [37] proposed ReSC-net, introducing a channel attention-enhanced residual module and spatial-channel attention to mitigate overfitting and optimize feature extraction for improved HSI classification.

It should be noted that the manual design of these deep network models demands extensive experience and knowledge in the relevant fields, making the process labor-intensive and time-consuming. To address these challenges, recent research has focused on automating the design process using NAS [38], [39], [40], [41]. This approach effectively uses the designed operators to search for a suitable network, reducing the manual workload to some extent. The early NAS methods mainly adopted discrete search strategies, including reinforcement learning [42] and evolutionary algorithms (EAs) [43]. These methods require the candidate architectures to be trained and evaluated at each step and fed back to the training network. In contrast, the contemporary NAS techniques often use one-shot NAS, which involves building a super-network covering all potential architectures [44]. The architecture variables within this framework are represented by flexible structural parameters, allowing optimization through continuous strategies such as gradient-based algorithms, and the final architecture comes from the convergence result of the search network. To date, several HSI classification methods that utilize one-shot NAS have been proposed [45], [46], [47], [48], [49].

However, in the process of constructing the search space, whether directly using the network architecture search

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operator in traditional vision tasks or the redesigned HSI classification operator, the candidate operations still have problems such as overemphasizing the spatial texture features and weakening the spectral information features under limited training samples, resulting in reduced classification accuracy. When encountering hyperspectral datasets with unique characteristics, candidate operations often cannot obtain a satisfactory search accuracy. In terms of search strategy, the differentiable gradient search method adopts the strategy of directly deriving the cell topological order from the operation weight, resulting in poor topological sorting correctness and difficulty in searching for the optimal solution.

In this article, in order to alleviate the above issues from the perspective of search space and search strategy, we propose a lightweight multiscale and multiattention NAS method based on decoupled operation search and topology search. Specifically, based on the characteristics of hyperspectral remote sensing images, we designed a new lightweight hyperspectral classification operator to ensure full utilization of the different receptive field features within the operator. In terms of search strategy, we decoupled the operation search from the topology search to maintain the correctness of the topological rating of the candidate edges. The main contributions of this work can be summarized as follows.

1) *New Lightweight Search Space Operators for HSI Classification:* In the proposed methodology, to comprehensively extract the spatial–spectral information from HSIs, a series of lightweight multiscale stacked convolutions have been designed. These convolutional structures replace the DSC and dilated convolution in differentiable architecture search (DARTS), thereby facilitating the extraction of texture features of objects across various scales.
2) *Feature Enhancement Under Multiple Attention Mechanisms:* In the design of the operator, spatial attention and spectral attention mechanisms are introduced to assign weights to the different spatial pixels and channel features, respectively, thereby enhancing the importance of the key spatial–spectral information. To achieve the fusion of features across different scales, a scale attention mechanism is incorporated to weight the feature maps at each scale, mitigating the issue of decreased classification accuracy caused by the differences in the scales of objects.
3) *Decoupling Strategy of Operation Search and Topology Search:* To identify a more suitable network architecture for HSI classification, this method introduces the concept of topological order. It decouples the topological representation from the search process, rectifies the inconsistencies in the topological ratings of each candidate operation within the target network, and thereby optimizes the network structure.

The remainder of this article is organized as follows. Section II introduces the relevant research background of lightweight and network structure search for HSI classification. Section III introduces the implementation details of L3M. Section IV provides the experimental results and related analysis of the model. Finally, Section V summarizes the whole article.

II. Related Research Background

A. Lightweight Operators and Methods for HSI Classification

Lightweight models aim to minimize the number of parameters and computational complexity while maintaining a high classification accuracy. Consequently, research has increasingly focused on lightweight structures and methods, gradually applying them to HSI classification. Significant progress has been made with lightweight operators in traditional vision tasks. Lightweight deep CNN (LDCNN) technologies, such as MobileNetv1 [50], MobileNetv2 [51], ShuffleNet [52], EfficientNet [53], and GhostNet [54], can achieve efficient operation on mobile devices by reducing the parameters and computation. For instance, MobileNetv1 significantly reduces the computational complexity through the use of DSC, while its successor—MobileNetv2—introduces an inverse residual module with a linear bottleneck to fully extract features by compressing and expanding the feature maps at different stages. ShuffleNet further optimizes the network structure via channel shuffling and pointwise group convolution (PG-Conv), and GhostNet employs phantom convolution to generate numerous feature layers through simple linear transformations, thereby reducing the computational cost. These lightweight operators have demonstrated strong performances in downstream tasks such as image classification, object detection, and image segmentation. Simultaneously, these techniques have been extensively applied to HSI classification, following appropriate modifications.

In terms of the lightweight models, substantial research has been conducted in both traditional vision tasks and hyperspectral remote sensing vision tasks. Quantization technology reduces the model storage and computational requirements by converting the model parameters and activation values from high precision to lower precision [55], [56]. Low-rank decomposition methods can exploit the linear or nonlinear correlations between data to compress the original high-dimensional data into a low-rank representation, thereby exploring the spectral and spatial dependencies [57]. Pruning technology decreases model size and computation by eliminating unimportant parameters or neurons [58]. Knowledge distillation technology maintains model performance while reducing size by training a smaller student model to mimic the output of a larger teacher model.

B. NAS for HSI Classification

In the domain of HSI classification, extensive research has been conducted on NAS methods that employ various strategies and on the design of NAS spaces tailored to contemporary tasks. Lu et al. [59] proposed a block-based evolutionary CNN (EBCNN) to automatically search for the best architecture based on a genetic algorithm (GA). Wan and Zhong [60] modeled the search problem as a multiobjective optimization problem and introduced a multiobjective EA to construct an HSI lightweight precision classification network (LPCN). A significant advancement of NAS in HSI

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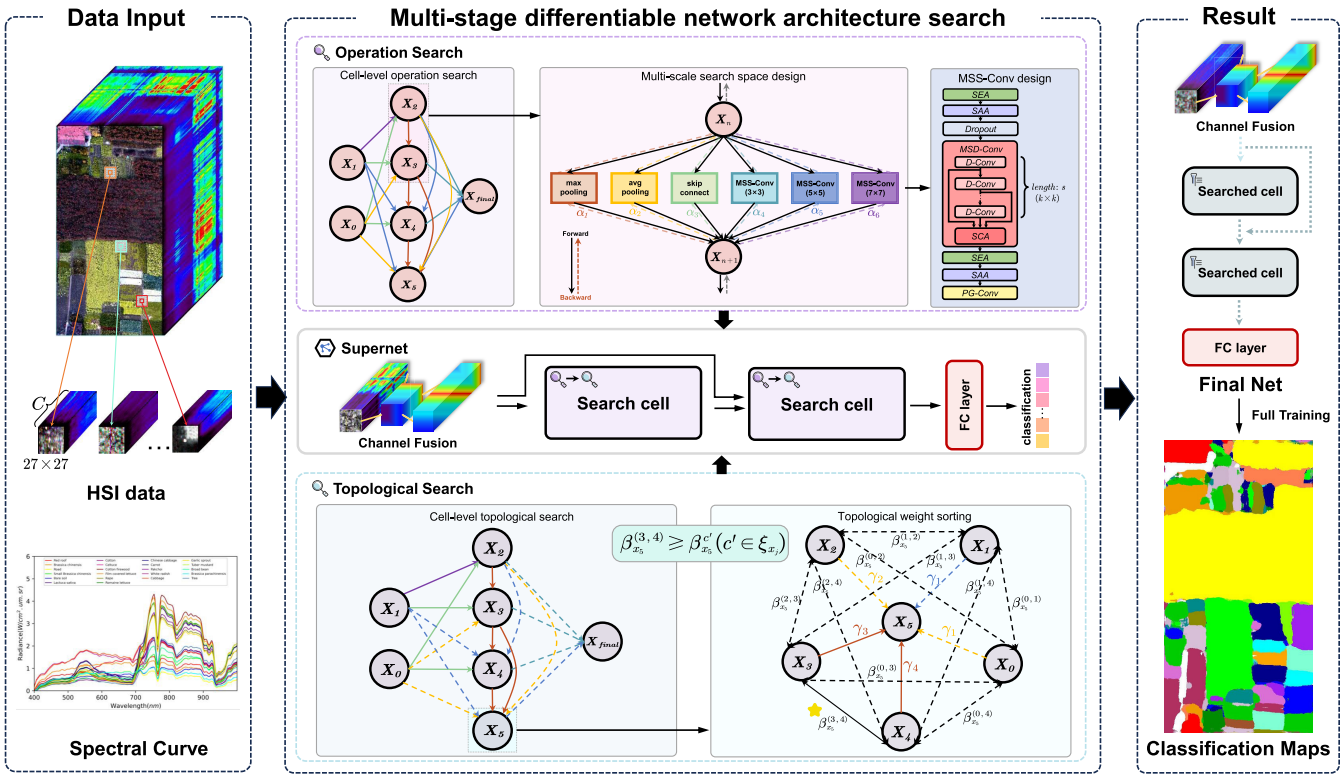


Fig. 1. Overall framework of the proposed L3M method.

classification was the introduction of DARTS. Chen et al. [61] first applied DARTS to HSI classification and proposed an Auto-CNN model. Subsequently, Zhang et al. [49] developed a 3-D asymmetric NAS (3D-ANAS) model that utilizes a 3-D asymmetric convolutional search space based on DARTS. This model not only significantly improves the classification performance, but also increases the model parameters and computational requirements, leading to potential overfitting problems. Cao et al. [45] proposed a novel lightweight multiscale NAS method combined with spatial–spectral attention. They developed an efficient search space that aims to minimize model parameters while enhancing the fusion of spectral and spatial features. Wang et al. [47] proposed a hyper kernel NAS method that directly generates architecture parameters using a specially designed hyper kernel.

III. Proposed L3M

In this article, we propose a lightweight network architecture search method predicated on a novel search space and an enhanced search strategy. The comprehensive search architecture is depicted in Fig. 1. This section commences with an in-depth discussion of the proposed framework, primarily elucidating the proposed hyperspectral classification operator that constitutes the search space, followed by an exposition of the improved search strategy.

A. Overall Framework of the L3M

The NAS framework can be systematically divided into three main components: the input of the HSI training set and validation set, the NAS process through training the supernetwork, and the output of the final network and classification results. HSI classification is analogous to the objective of semantic segmentation in traditional vision tasks, as it aims to accurately classify each pixel in the HSI. To leverage the spatial local feature extraction capability of the convolution operation, we employ a patch-based dataset partitioning strategy during the data preparation stage. Specifically, given a hyperspectral dataset denoted as $D \in \mathbf{R}^{H_{\text{whole}} \times W_{\text{whole}} \times C}$, the target pixel to be classified is taken as the center point to define the neighborhood cube $I \in \mathbf{R}^{H \times W \times C}$, where C represents the number of input channels and $H \times W$ represents the spatial dimension size of the cube. The label corresponding to the current data cube is the center pixel label. The network architecture search is primarily divided into operation search and topology search. The input of the topology search is the weight output from the operation search, which updates the operation weight and edge combination weight, respectively, and the output is the final network and the result of the two-stage search. Consequently, the classification result is generated by the final model trained by the network.

B. Search Space Design

NAS-based methods necessitate a predefined search space to conduct search operations. To ensure that the resultant network possesses an efficient architecture with a minimal number of parameters, we devised a lightweight search space, as illustrated in Table I. This search space encompasses both parameterized and nonparameterized operations. The nonparameterized operations adhere to those employed in

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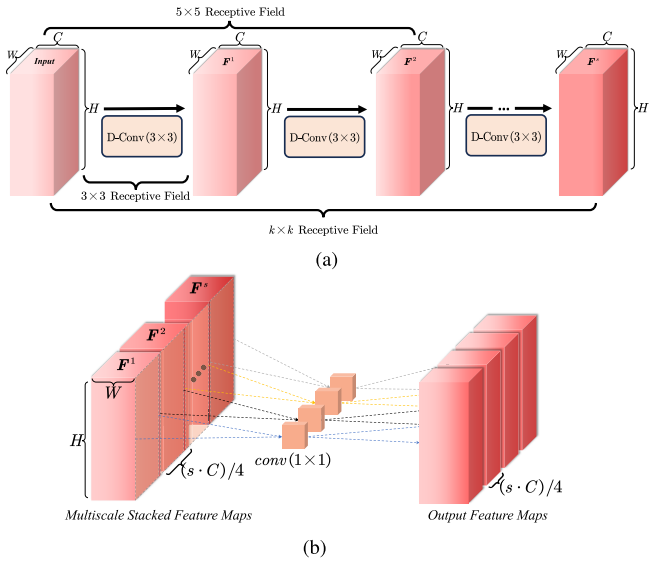


Fig. 2. Convolution operator. (a) MSD-Conv. (b) Point grouping convolution.

the DARTS framework. Specifically, these include skip connections, which effectively mitigate gradient vanishing. The maximum pooling and average pooling operations, which maintain the feature map size, efficiently eliminate redundant information, enhance the model's nonlinearity, and facilitate the extraction of a substantial number of features with minimal operations, thereby augmenting the model's information extraction capability. The parameterized operations consist of a multiscale and multiattention stacked convolution (MSS-Conv) operator. The detailed structure of this operator is elucidated in Sections III-B1 and III-B2.

1) Multiscale Stacked DW-Conv: In contemporary research, the majority of the parameterized operations in NAS are constructed based on convolution. Previous studies have primarily utilized traditional convolution, DSC, and dilated convolution as alternative operations. Each type of convolution possesses distinct feature extraction capabilities and is suited to different practical scenarios. However, with the continuous stacking of network cells in NAS, it is imperative to reduce the number of network parameters and computational cost while maintaining the network scale. Inspired by LMSS-NAS [45], we aim to minimize the structural parameters and computational complexity of the convolution kernel while preserving its feature extraction capability and receptive field size. To achieve this, we perform equivalent operations on larger scale filters. As depicted in Fig. 2(a), 5×5 and 7×7 convolutions can be equivalently replaced by multiple groups of stacked 3×3 convolutions, and 3×3 convolutions can be equivalently replaced by 1×3 and 3×1 strip convolutions. During this process, large receptive field convolutions naturally generate multiple levels of feature maps. To fully utilize features of different scales, we retain the 3×3 feature maps at each step for multiscale feature fusion. In addition, to further reduce the number of parameters, each 3×3 convolution operation employs D-Conv to extract spatial feature information, while the centralized extraction and enhancement of spectral information is delegated to the subsequent spectral attention mechanism and point convolution processing. In the final stage, local spectral feature extraction is performed to reduce the spectral feature redundancy and retain local band information. The structure of PG-Conv is shown in Fig. 2(b). This approach enables the network to learn the ground object information reflected by different bands. The function of the multiscale stacked DW-Conv (MSD-Conv) component can be expressed as follows:

$$\text{D-Conv} = \boldsymbol{f}_{\text{BN}}(\text{conv}(3 \times 1)(\text{conv}(1 \times 3)(\delta(F)))) \tag{1}$$

$$\text{MSD-Conv}(k \times k) = \text{concat}\big(\text{D-Conv}^1(F) \quad \text{D-Conv}^2(F), \ldots, \text{D-Conv}^s(F)\big). \tag{2}$$

Given that $k = 2s + 1$, where k denotes the cardinality and represents the minimum size of the convolution kernel, which is 3×3, and $s \in \{1, 2, 3\}$. $\text{conv}(w \times h)$ represents DW-Conv, $\delta(\cdot)$ indicates the ReLU activation function, and dc^s means that s dc operations are performed.

The parameter count (Params) and floating-point operations (FLOPs) in the overall operator are primarily determined by the design of the internal convolution operations. The parameter count of a single convolution kernel is given by $P = C_{\text{in}} \times k \times k \times C_{\text{out}}$, and the FLOPs can be expressed as $F = P \times H \times W$. By employing MSD-Convs, the Params and FLOPs of conventional convolutions with kernel sizes $k = 3, 5, 7$ are transformed into fractions of their original values: $(2 \times 1 \times 3)/(3 \times 3)$, $(2 \times 2 \times 1 \times 3)/(5 \times 5)$, and $(3 \times 2 \times 1 \times 3)/(7 \times 7)$, respectively, leading to a reduction of approximately 33.3%∼63.2%. Consequently, a higher proportion of large-scale convolutions in the searched architecture generally results in a greater reduction in model Params and FLOPs.

2) Multiple Attention Mechanisms: To enhance the capability of extracting contextual information from the channel and spatial features of the designed operator, lightweight spatial attention and spectral attention mechanisms are employed to amplify the significance of crucial channels and pixels. As illustrated in Fig. 3, the spatial attention mechanism initially compresses information along the channel dimension by computing channel-wise average and maximum values. Subsequently, the refined output feature map is obtained through operations such as convolution. In contrast, the spectral attention mechanism compresses the input feature map along the spatial dimension using global average pooling and global maximum pooling, resulting in a 1-D feature vector. This 1-D feature vector more effectively captures and reallocates the relative importance across channels by utilizing the ReLU activation function and employing compression and expansion operations. In addition, to emphasize the spectral characteristics of the central pixel as the classification target, the spectral weight of the central point is integrated with the afore-mentioned 1-D feature vector, serving as a reference for channel weight distribution.

Aiming to fully leverage the diverse-scale stacking operators generated by a single stacked convolution operator, a scale attention mechanism is introduced, inspired by the S3ANet

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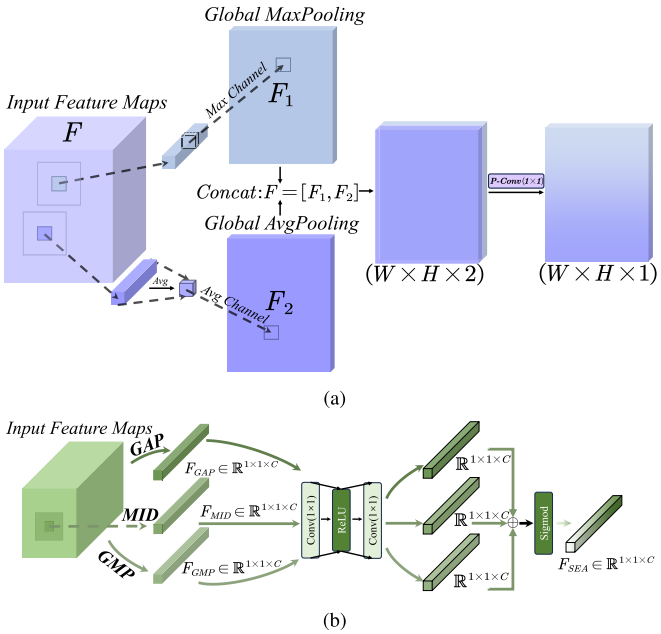


Fig. 3. Illustration of the attention mechanisms used by the operator. (a) Spatial attention mechanism. (b) Spectral attention mechanism.

[22]. Analogous to spatial attention and spectral attention, scale attention adaptively weights the scale information, effectively addressing the scale difference problem. As illustrated in Fig. 4, the multiscale stacked feature maps are aggregated through element-wise summation to integrate features from different scales. The statistical features of the fused channel S are then derived using global average pooling and global maximum pooling. To minimize network parameters, the feature vector S is further compressed, and the compressed feature vector undergoes matrix operations with an adaptive scale tensor to compute the absolute weight for each scale. After being processed by the softmax function, these weights are subsequently multiplied with the initial multiscale stacked feature maps to produce the scale-weighted feature map X.

Finally, to integrate channel information and enhance the model's nonlinear representation capability, the feature map, enriched by the multiple attention mechanisms, is processed using PG-Conv. Unlike traditional pointwise convolution, this approach maintains the spatial structure while effectively fusing the multiple attention weights, and consequently reduces the model complexity and the number of parameters. The strategy for utilizing multiple attention mechanisms is outlined as follows:

$$\mathrm{M_{SAA}}(F) = \sigma(\mathrm{conv}_{7\times7}(\mathrm{concat}(\mathrm{GAP}(F), \mathrm{GMP}(F)))) \quad (3)$$

$$\mathrm{M_{SEA}}(F) = \sigma(\mathrm{MLP}(\mathrm{GAP}(F)) + \mathrm{MLP}(\mathrm{GMP}(F)) + \mathrm{MID}(F)) \quad (4)$$

$$\mathrm{M_{SCA}}(F) = \phi\big(\big(\delta\big(\boldsymbol{f}_{\mathrm{BN}}(\mathrm{MLP}(\mathrm{GAP}(F)))\big) + \delta\big(\boldsymbol{f}_{\mathrm{BN}}(\mathrm{MLP}(\mathrm{GMP}(F)))\big)\big) \cdot W\big) \quad (5)$$

where $\sigma(\cdot)$ indicates the sigmoid activation function and $\phi(\cdot)$ indicates the softmax activation function. Let $W \in \mathbf{R}^{s\times b\times b/\gamma}$, where b denotes the number of channels in the input multiscale fusion feature map, and γ represents the compression ratio, which is defaulted to 16. As shown in the "MSS-Conv design" module in Fig. 1, the MSS-Conv operator is organically made up of MSD-Conv and the multiple attention mechanisms.

The input feature map is initially enhanced through spectral–spatial attention mechanisms. This enhanced feature map then undergoes convolution via MSD-Conv to capture multiscale spatial information. The features retained at each scale are then reweighted to match the current object size. Subsequently, attention mechanisms in both the spectral and spatial dimensions are applied to further adjust the feature map based on the channel and spatial position importance. This process ensures that the final feature map emphasizes the critical regions and channels, thereby improving the overall output quality. The operations included in the overall search space are listed in Table I. Where o_i^n represents a parameterless operation, and o_i^y represents a parameterized operation. Dropout (DR) is introduced in the parameterized operation to reduce overfitting and improve the generalization ability of the operator.

TABLE I
DETAILED COMPONENTS OF ALL CANDIDATE OPERATIONS IN THE SEARCH SPACE

ID	Operations Name	Detailed Components
o_1^n	$skip_connect$	$f(x) = x$
o_2^n	$avg_pooling$	$AvgPooling2d$
o_3^n	$max_pooling$	$MaxPooling2d$
o_1^y	MSS-Conv(3×3)	$\mathrm{M}_{SAA}-\mathrm{M}_{SEA}-DR-$MSD-Conv($3\times3$) $-\mathrm{M}_{SCA}-\mathrm{M}_{SAA}-\mathrm{M}_{SEA}$
o_2^y	MSS-Conv(5×5)	$\mathrm{M}_{SAA}-\mathrm{M}_{SEA}-DR-$MSD-Conv($5\times5$) $-\mathrm{M}_{SCA}-\mathrm{M}_{SAA}-\mathrm{M}_{SEA}$
o_3^y	MSS-Conv(7×7)	$\mathrm{M}_{SAA}-\mathrm{M}_{SEA}-DR-$MSD-Conv($7\times7$) $-\mathrm{M}_{SCA}-\mathrm{M}_{SAA}-\mathrm{M}_{SEA}$

C. Decoupled Operation and Topology Search

Based on the DARTS search paradigm, the proposed method addresses the limitations of optimal unit topology discovery by incorporating decoupled operation and topology searches. As illustrated in Fig. 1, the search process is divided into operation search and topology search phases. During the operation search phase, the optimal operations for each cell are identified through automated search, with the performance evaluated based on parallel weighting. In the topology search phase, edge weights are assigned according to predefined combination weights. Sections III-C1 and III-C2 detail the construction of the topology search space and elucidate the integration of the operation search strategy into the topology search, facilitating seamless connection across multiple search stages.

1) Establishment of Topology Search Space: Adhering to the default search strategy of DARTS, the importance of an

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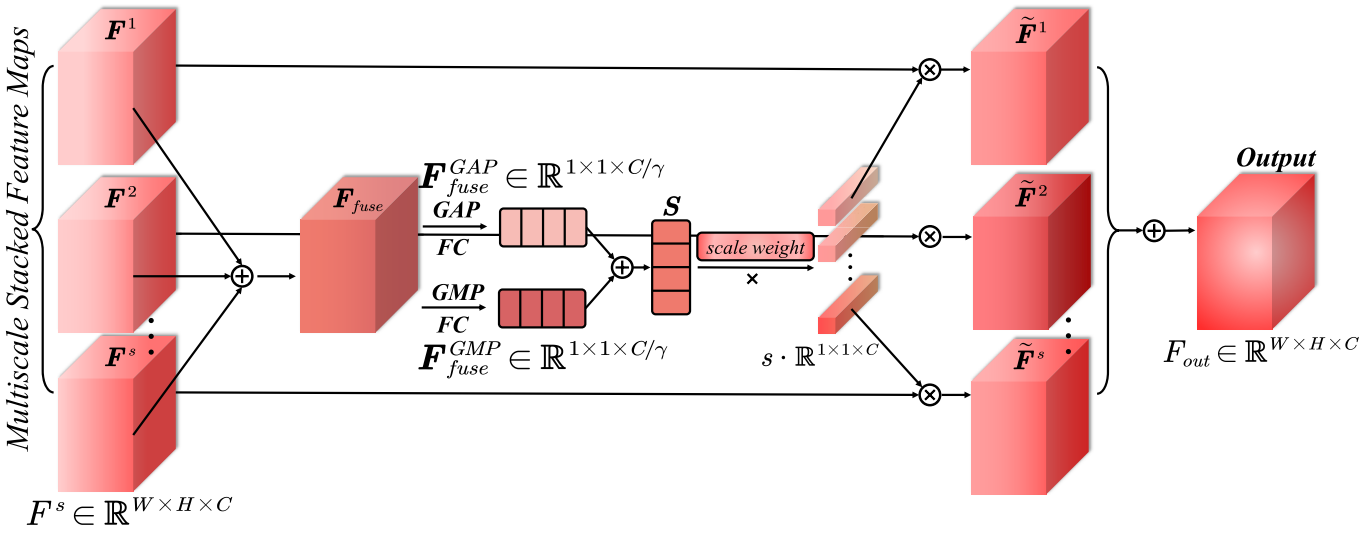


Fig. 4. Network structure of the scale attention mechanism. The input is a multiscale stacked feature map, and the output is a scale-weighted fusion feature map.

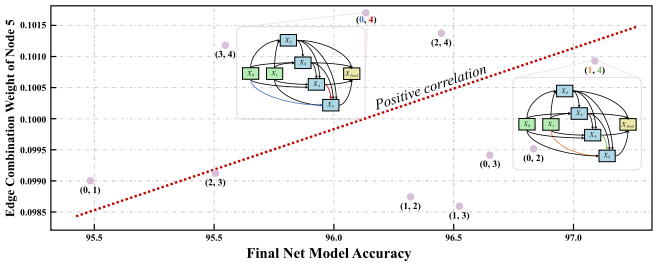


Fig. 5. Correlation analysis of edge combination weight and classification accuracy. The vertical axis represents the edge combination of the fifth node. Without topology search, the topological weight lacks a positive correlation with accuracy, resulting in suboptimal selections.

edge combination is determined by summing the weights of the operations with the highest values on the two selected edges. As depicted in Fig. 5, while a higher combination weight signifies the adaptability of a candidate operation to a particular edge, it does not necessarily indicate its optimality within the edge topology, postcombination. The lack of a clear ranking correlation between the edge combination importance and search result accuracy underscores that the default screening strategy is insufficient for converging to the optimal edge combination topology. To this end, a topology search space is introduced to optimize the search strategy from the structural level [62].

To formulate the search space, the node x_j is connected to all its predecessor nodes through the edge (i, j). The number of the preceding node is $i \in \{i|i < j\}$. Based on DARTS candidate edge screening strategy, which limits intermediate nodes to two edges, the topology search space ξ_{x_j} for x_j is constructed as the complete set of pairwise combinations of all incoming edges

$$\xi_{x_j} = \{\langle (i_1, j), (i_2, j)\rangle \mid 0 < i_1 < i_2 < j\}. \tag{6}$$

For each node x_j, in order to maintain a smooth distribution of the combined weights in the topological space of the point, the annealing temperature T_β is introduced during the softmax operation, which is specifically defined as

$$\beta_{x_j}^c = \frac{\exp\left(\beta_c'^{x_j}/T_\beta\right)}{\sum_{c'\in\xi_{x_j}} \exp\left(\beta_{c'}'^{x_j}/T_\beta\right)} \tag{7}$$

where $\beta_{x_j}^c$ denotes the normalized probability that selects the edge combination c as its incoming edge. Annealing is performed using an exponential schedule θ^t, starting from an initial temperature T_0 and decaying as the number of training rounds increases. Since the operation search yields the weight of only a single edge, it is essential to compute the combined features of each edge to adjust the operation weight. To achieve this, the weights of all combinations containing the current edge e_j^i are directly extracted and summed, which can be formalized as

$$\gamma^{(i,j)} = \sum_{c\in\xi_{xj},(i,j)\in c} \beta_{x_j}^c \tag{8}$$

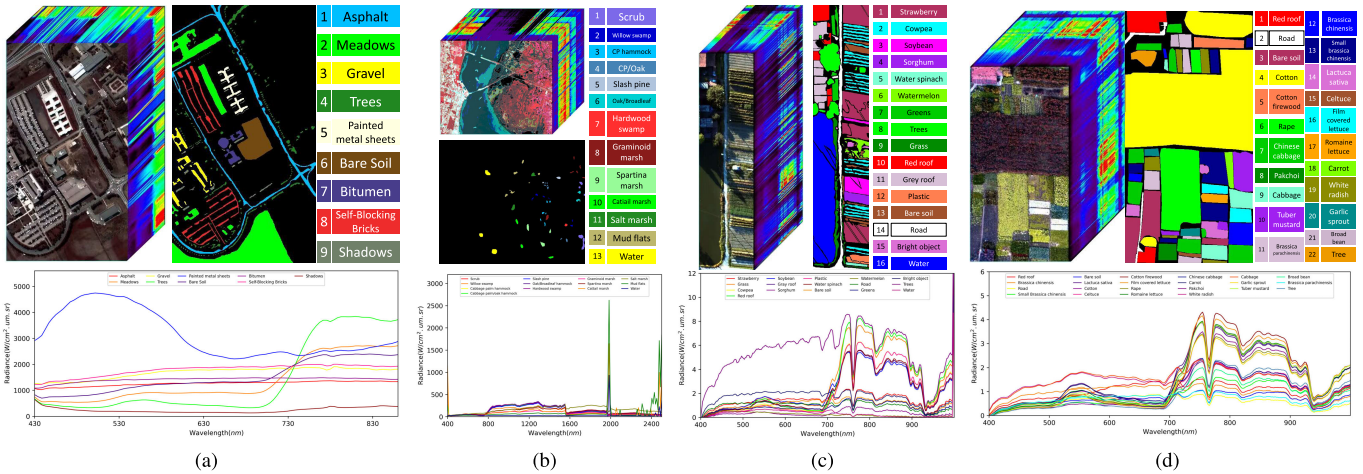
where $\gamma^{(i,j)}$ is the topological weight of each edge. To mitigate numerical errors arising from being too small during the subsequent fusion with the operation weight, weight aggregation is directly performed through addition and summation.

2) Integration of the Search Strategy: The previous DARTS-based operation weight solving strategy can be easily incorporated into the topology search strategy. According to the design principle of DARTS, each edge (i, j) will be weighted by an operation weight $\alpha^{(i,j)}$, which can be defined as

$$\overline{o}^{(i,j)}(x) = \sum_{o\in\mathcal{O}} \alpha_o^{(i,j)} o^{(i,j)}(x_i) \tag{9}$$

where $o(x) \in \mathcal{O}$ and $\mathcal{O}$ is the operation search space containing six operations in Table I. At this stage, to combine the topological weight coefficients, the original hybrid

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(a) (b) (c) (d)

Fig. 6. Four HSI classification datasets. (a) Pavia University (PU), (b) Kennedy Space Center (KSC), (c) WHU-Hi-Hanchuan, and (d) WHU-Hi-Honghu. The false-color images, ground-truth maps, and spectral curves of each dataset are shown.

operation is decoupled and only the operation with the highest weight on edge (i, j) is retained

$$o^{(i,j)} = \arg\max_{o \in \mathcal{O}} \alpha_o^{(i,j)}. \tag{10}$$

In simple terms, the mixing operation $\overline{o}^{(i,j)}$ in DARTS is replaced by the maximum weight operation $o^{(i,j)}$.We weight all incoming edges of node x_j by $\gamma^{(i,j)}$ for

$$x_j = \sum_{i<j} \gamma^{(i,j)} \cdot o^{(i,j)}(x_i). \tag{11}$$

The input of the intermediate node x_j is derived from the output of the predecessor node x_i. For the use of the optimization update strategy, the bilevel optimization strategy employed by DARTS is discarded, and one-level optimization is used to achieve stability in the search convergence. In the topology search phase, deriving the topological weights of a single edge by using the edge combination weights greatly reduces the operations on each edge and eliminates the risk of overfitting. The formulaic representation of using one-level optimization to update the network weights w and topological weights β is as follows:

$$\begin{aligned} w_t &= w_{t-1} - \eta_t \partial_w L_{\text{train}}(w_{t-1}, \beta_{t-1}) \\ \beta_t &= w_{t-1} - \delta_t \partial_w L_{\text{train}}(w_{t-1}, \beta_{t-1}) \end{aligned} \tag{12}$$

where η_t is the network weight learning rate and δ_t is the topology weight learning rate.

IV. Experiments and Analysis

In this section, we aim to validate the effectiveness of the proposed method in classifying hyperspectral datasets with varying characteristics. To achieve this, we evaluated L3M on four real-world hyperspectral datasets. The subsections are organized as follows. First, we provide a brief overview of the datasets. Next, we detail the implementation procedures and parameter configurations used in the model construction process. Subsequently, we assess the model performance by comparing L3M with other lightweight models and NAS-based hyperspectral imaging methods, using key metrics such as the overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa). To demonstrate that L3M satisfies the lightweight requirements, we also present a comparison of the model efficiency, focusing on training parameters (Params), FLOPs, GPU memory usage, and inference speed.

We evaluated the impact of the proposed lightweight operator and search strategy through ablation studies. To further validate the appropriateness of the search space design, we explored the potential for improving the architecture via hyperparameter analysis. Finally, by introducing the uniform distribution indicator of the data, we analyzed the effect of uniformity of the training data on the performance of the lightweight model.

A. Datasets and Implementation Details

In the experimental design, we selected two widely recognized HSI classification benchmarks. The first dataset—the Pavia University dataset—was collected in 2001 using the Reflective Optics System Imaging Spectrometer (ROSIS) sensor in Pavia, northern Italy. The second dataset—the Kennedy Space Center (KSC) dataset—was captured in 1996 by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor at the KSC in Florida, USA. To further demonstrate the adaptability of the proposed search model to hyperspectral datasets with a high spatial resolution (H2), we additionally utilized the WHU-Hi-Hanchuan and WHU-Hi-Honghu datasets, which were provided by the Remote Sensing Intelligent Data Extraction and Analysis (RSIDEA) group at Wuhan University in China [63]. These datasets were collected in Hubei province, China, in 2018, using an autonomous aerial vehicle (AAV)-borne hyperspectral system. The WHU-Hi datasets pose significant challenges due to the large variety of crop types, the vast number of pixels, and the high spatial resolution of the imagery, which leads to diverse sub-class scales of different land cover types and often results in notable scale variations. A detailed summary of the datasets can be found in Fig. 6 and Table II.

To evaluate the feature learning capability of the proposed method, we followed the sample selection strategy

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TABLE II
SPECIFIC ATTRIBUTES OF THE DATASETS AND DISTRIBUTION RATIO OF EXPERIMENTAL DATA

Pavia University		KSC		WHU-Hi-HanChuan		WHU-Hi-HongHu	
spectral bands	103	spectral bands	176	spectral bands	274	spectral bands	270
Spatial resolution	18m	Spatial resolution	1.3m	Spatial resolution	0.109m	Spatial resolution	0.043m
image size	610×340	image size	512×614	image size	1217×303	image size	940×475
wavelength range	0.43-0.86μm	wavelength range	0.4-2.5μm	wavelength range	0.4-1.0μm	wavelength range	0.4-1.0μm
Total	42776	Total	5211	Total	257530	Total	417567
Training samples	180	Training samples	260	Training samples	320	Training samples	440
Validation samples	90	Validation samples	130	Validation samples	160	Validation samples	220

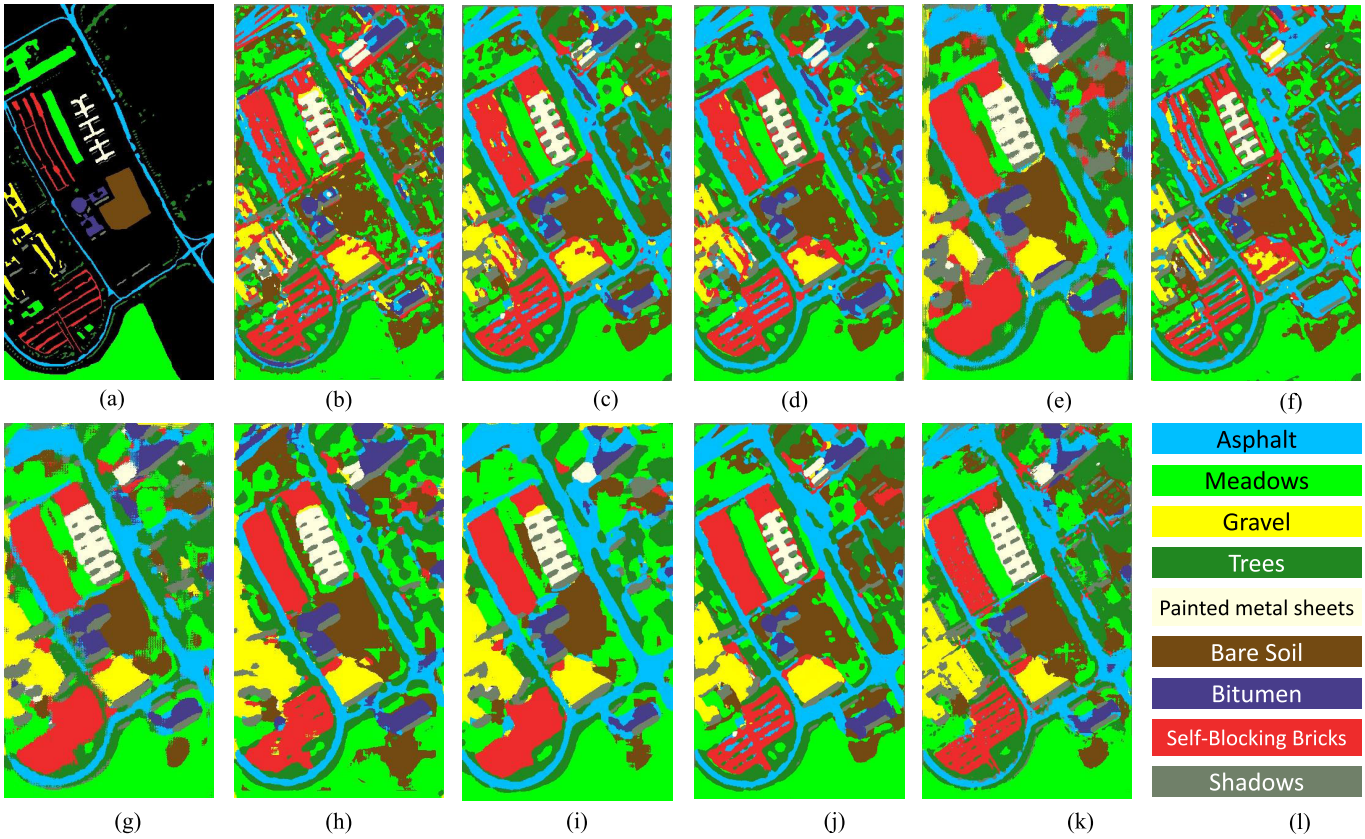


Fig. 7. Classification results of the Pavia University dataset. (a) Ground Truth. (b) A^{2}S^{2}K-ResNet. (c) CLOLN. (d) CLOLN-Ghost. (e) LS^{2}CM. (f) LMAFNet. (g) 3D-ANAS. (h) 3D-AT-CNN. (i) LMSS-NAS. (j) LPCN. (k) L3M. (l) Legend.

from lightweight multiscale spatial-spectral attention NAS (LMSS-NAS) [45], using 20 samples for the training set and ten samples for the validation set. The specific proportions of the training and evaluation samples across each dataset are detailed in Table II. To ensure fairness in the comparative analysis and obtain a reliable average model performance, each experiment was repeated ten times, with the mean value of each metric recorded as the final result.

In this experiment, to address the spatial feature extraction requirements of the proposed operator, we configured the hyperspectral cube window size to 27×27 and set the batch size for both the training and inference to 32. The two-stage search process was optimized using the Adam optimizer, with a default weight decay of 0.0003. The learning rates were set to 0.004 and 0.0003, respectively. For the two-stage super-network architecture search, 100 and 400 iterations were conducted in each stage to ensure convergence. During the final network optimization, 500 iterations were performed. To ensure consistency in the experimental comparisons, we retained the original hyperparameters specified by the patch-based methods without intentionally aligning them with the other experimental conditions. The key hyperparameter settings for each model are summarized in Table III.

Unlike the previous differentiable NAS methods, the proposed L3M method focuses exclusively on searching for reduction cells across the datasets. The final network is composed of two connected reduction cells. This approach leverages the tendency of reduction cells to favor convolution operations for extracting deep features while simultaneously eliminating redundant information and reducing the parameter

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TABLE III
TRAINING SETTINGS OF THE MODELS FOR COMPARISON

Model	A^2S^2K-ResNet	CLOLN	CLOLN-Ghost	LS^2CM	LMAFNet
Epoch	200	300	300	500	4000
Patch	9×9	11×11	11×11	11×11	9×9
Optimizer	Adam	Adam/SGD	Adam/Adam	SGD	Adam
Lr	0.01	0.001	0.01	0.04	0.001
Model	3D-ANAs*	3D-AT-CNN*	LMSS-NAS*	LPCN*	L3N*
Epoch	100/600	100/200	100/500	200/500	500/500
Patch	32×32	32×32	27×27	32×32	27×27
Optimizer	Adam	Adam/Adam	Adam/Adam	Adam/Adam	Adam/Adam
Lr	0.001	0.004	0.004	0.001	0.001

The models marked with * represent NAS-based methods.

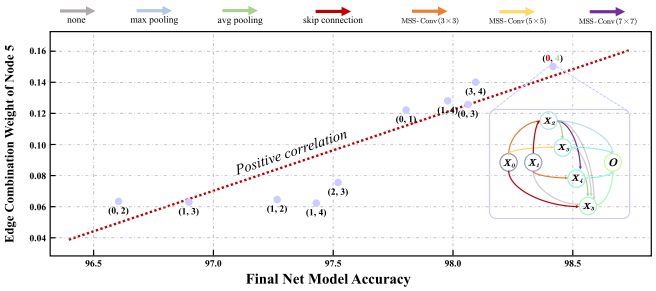


Fig. 8. Optimal structure of cells based on L3M for the Pavia University dataset, and the topological rating analysis.

count. The differences in the architectures discovered across the datasets highlight the adaptability of the search strategy to various data characteristics. Notably, the parameterized operations in the reduction cells often prioritize deriving feature maps from earlier blocks, which facilitates the rapid integration of spatial and spectral information, enabling the network to capture distinguishable features at an earlier stage.

B. Classification Results

To evaluate the effectiveness of the proposed L3M model, we conducted a comparative analysis with eight state-of-the-art models: the A^2S^2K-ResNet [30], the channel-layer-oriented lightweight spectral–spatial network (CLOLN), CLOLN-Ghost [64], the lightweight spectral–spatial convolution module (LS^2CM) [65], NAS-guided lightweight multiscale attention fusion network (LMAFNet) [32], 3D-ANAS [49], the automatic search version of 3-D CNN (3D-AT-CNN) [61], LMSS-NAS [45], and the lightweight precise classification network (LPCN) [60]. Among these models, the first five models are manually designed, while the latter four utilize NAS methodologies. Notably, CLOLN, LS^2CM, LMAFNet, LMSS-NAS, and LPCN incorporate lightweight design principles in their construction.

We assessed the overall classification performance of each model across the various datasets using three key metrics: OA, AA, and the Kappa. The classification accuracies, along with the OA, AA, and Kappa values for each category on the Pavia University, KSC, WHU-Hi-Hanchuan, and WHU-Hi-Honghu datasets, are detailed in Tables IV–VII. In addition, the classification results for each model across the entire HSIs are visually presented in Figs. 7, 9, 11, and 13. This comprehensive evaluation enables a thorough understanding of the proposed model's performance, relative to the existing methodologies.

1) Experiment With the Pavia University Dataset: The search result for the Pavia University dataset is shown in Fig. 8. For the Pavia University dataset, the proposed L3M model outperforms all the others across the three overall indicators. As shown in Table V, L3M not only achieves the best classification results but also delivers a near-optimal performance in each category. When compared to the comparison models of LMSS-NAS and 3D-ANAS, L3M notably enhances the texture features of each patch during the classification process. This refinement is achieved while maintaining overall smoothness, resulting in a more precise full-image classification effect. This underscores the effectiveness of the proposed L3M model in balancing detail and coherence in the classification outcome. Furthermore, to validate the effectiveness of the topology search, we present a scatter plot illustrating the relationship between the edge combination weights of the fifth node and the corresponding classification accuracy. As depicted in Fig. 8, there is a clear positive correlation between the topological weight and the classification accuracy of the final network.

2) Experiment With the KSC Dataset: For the KSC dataset, the search result is shown in Fig. 10. In terms of accuracy, the proposed L3M model continues to demonstrate a superior classification performance among the labeled samples, which is consistent with the previous experiment. Analyzing the classification result graphs in Fig. 9 reveals that the L3M model not only accurately identifies the sample points within the specified area, but also effectively delineates the unlabeled features in the image. Notably, L3M captures distinct texture features, such as bridges, thereby illustrating its capability to recognize and characterize complex structures within the dataset. This reinforces the model's effectiveness in both precise identification and broader contextual understanding. We also verified the necessity of correct topology ranking when searching for the optimal model on the KSC dataset, as shown in Fig. 10.

3) Experiment With the WHU-Hi-Hanchuan Dataset: For the WHU-Hi-Hanchuan dataset, the search result is shown in Fig. 12. Again, the proposed L3M model consistently outperforms the others, achieving the best results across all three quantitative indicators. A comparative analysis of the result graphs reveals that the L3M model excels in classifying local areas characterized by diverse landforms, particularly in the mixed regions of watermelon, bare soil, and bright soil located to the right of the dataset's center. In contrast, the other models fail to achieve effective local classification in this area, even under identical training conditions. This underscores the proposed model's superior capability in handling complex and heterogeneous landscapes. We further validated the importance of accurate topology ranking in the process of searching for the optimal model on the WHU-Hi-Hanchuan dataset, as shown in Fig. 12.

4) Experiment With the WHU-Hi-Honghu Dataset: For the WHU-Hi-Honghu dataset, the proposed L3M model not only achieves the highest classification accuracy but also maintains a relatively smooth classification boundary, in comparison to

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TABLE IV
EXPERIMENTAL RESULTS ON THE PU DATASET

Models	A^{2}S^{2}K-ResNet	CLOLN	CLOLN-Ghost	LS^{2}CM	LMAFNet	3D-ANAs*	3D-AT-CNN*	LMSS-NAS*	LPCN*	L3M*
1	84.12	97.71	99.08	53.06	96.15	93.21	92.24	97.60	98.45	**99.49**
2	95.67	98.21	95.93	70.36	90.71	99.10	94.72	**99.54**	99.10	99.51
3	87.23	97.57	88.47	80.82	97.01	**99.28**	96.11	94.61	96.42	97.75
4	91.45	96.01	94.94	78.60	96.28	**99.22**	85.27	95.28	96.28	96.17
5	93.89	99.32	95.48	98.16	99.77	**100.00**	97.12	98.64	98.98	**100.00**
6	85.30	82.70	83.69	73.50	92.89	97.36	97.47	89.66	92.45	**100.00**
7	96.78	95.64	54.73	92.79	98.00	**99.62**	95.14	89.09	93.12	98.69
8	89.12	95.24	92.91	40.54	89.34	87.35	96.84	**97.99**	93.67	94.93
9	97.45	75.63	83.61	82.44	**100.00**	98.69	83.62	95.74	95.88	93.19
OA(%)	89.04	94.93	91.53	68.38	93.10	96.53	95.16	96.78	97.16	**98.67**
AA(%)	85.17	93.11	87.65	74.48	95.57	96.31	94.72	95.35	96.04	**97.75**
Kappa	0.8530	0.9333	0.8884	0.6046	0.9097	0.9539	0.9251	0.9575	0.9636	**0.9824**

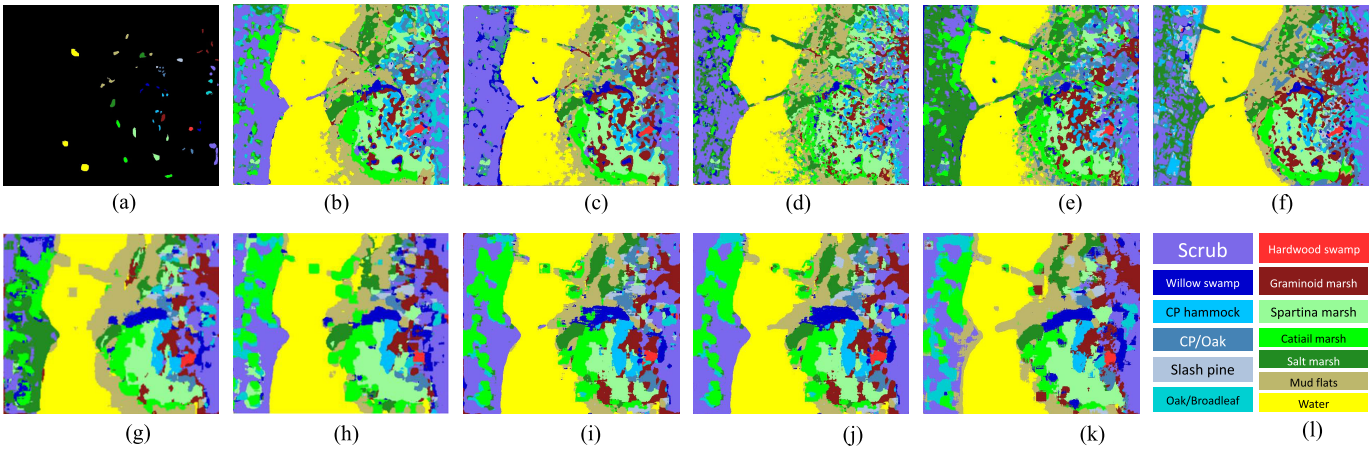


Fig. 9. Classification results of the KSC dataset. (a) Ground truth. (b) A^{2}S^{2}K-ResNet. (c) CLOLN. (d) CLOLN-Ghot. (e) LS^{2}CM. (f) LMAFNet. (g) 3D-ANAS. (h) 3D-AT-CNN. (i) LMSS-NAS. (j) LPCN. (k) L3M. (l) Legend.

TABLE V
EXPERIMENTAL RESULTS ON THE KSC DATASET

Models	A^{2}S^{2}K-ResNet	CLOLN	CLOLN-Ghost	LS^{2}CM	LMAFNet	3D-ANAs*	3D-AT-CNN*	LMSS-NAS*	LPCN*	L3M*
1	97.40	**100.00**	**100.00**	**100.00**	99.32	84.15	**100.00**	**100.00**	**100.00**	**100.00**
2	96.06	98.61	90.75	92.66	**100.00**	95.29	93.42	99.07	98.20	98.16
3	99.60	89.20	81.78	68.50	**100.00**	94.18	**100.00**	99.56	99.53	99.56
4	97.18	72.33	80.58	85.19	95.05	89.31	**99.53**	97.72	97.89	98.76
5	98.67	93.64	86.62	84.68	99.71	**100.00**	75.58	96.18	97.20	98.67
6	96.68	96.86	88.18	**100.00**	**100.00**	88.54	98.02	**100.00**	**100.00**	**100.00**
7	**100.00**	**100.00**	98.68	93.75	**100.00**	97.36	**100.00**	**100.00**	**100.00**	**100.00**
8	99.38	95.93	85.33	81.60	99.01	87.78	93.47	**99.75**	**99.75**	**99.75**
9	97.73	**100.00**	91.91	**100.00**	**100.00**	99.75	**100.00**	98.79	98.99	**100.00**
10	99.71	**100.00**	93.35	97.61	**100.00**	95.24	**100.00**	**100.00**	**100.00**	**100.00**
11	98.89	**100.00**	**100.00**	**100.00**	**100.00**	96.96	**100.00**	**100.00**	**100.00**	**100.00**
12	96.19	**100.00**	97.30	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
13	97.15	98.90	98.79	99.23	**100.00**	93.94	**100.00**	**100.00**	**100.00**	99.89
OA(%)	99.27	97.12	93.63	94.34	99.52	94.08	98.13	99.59	99.60	**99.79**
AA(%)	99.15	95.81	91.79	92.55	99.31	93.15	96.92	99.31	99.35	**99.65**
Kappa	0.9889	0.9677	0.9289	0.9367	0.9946	0.9345	0.9791	0.9954	0.9787	**0.9975**

the other classification models. This performance, combined with the results observed in the previous two complex scenarios, indicates that the proposed method demonstrates robust efficacy in generating discriminative classification maps. L3M effectively addresses both the traditional satellite hyperspectral datasets and datasets characterized by complex texture features and a greater variety of categories, highlighting its versatility and reliability in diverse applications. As illustrated in Figs. 8, 10, 12, and 14, the network structures identified through the search process across the four datasets demonstrate a clear tendency to apply parameterized operations on the original input, followed by the generation of feature maps at different granularities through the parameter-free operations. As demonstrated in Fig. 14, the final network's classification

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TABLE VI
EXPERIMENTAL RESULTS ON THE WHU-HI-HANCHUAN DATASET

Models	A^{2}S^{2}K-ResNet	CLOLN	CLOLN-Ghost	LS^{2}CM	LMAFNet	3D-ANAs*	3D-AT-CNN*	LMSS-NAS*	LPCN*	L3M*
1	92.96	95.99	92.04	94.08	93.48	96.21	96.22	96.84	95.56	**98.27**
2	87.38	89.66	95.34	92.19	79.58	**98.71**	95.59	99.27	93.15	94.92
3	**97.89**	80.80	63.76	96.10	90.23	86.34	82.71	97.28	82.98	85.22
4	93.74	87.86	76.49	90.39	99.40	83.08	90.98	95.54	**98.58**	91.80
5	75.96	48.65	33.92	49.45	**100.00**	93.83	68.09	96.92	75.53	77.43
6	93.55	45.63	25.03	**99.32**	83.74	76.59	80.65	66.64	81.28	84.98
7	47.81	60.28	71.13	81.58	**94.93**	85.82	71.53	69.41	90.82	76.22
8	66.34	93.77	78.95	70.29	74.46	80.76	**97.91**	92.30	75.60	94.15
9	74.63	76.69	28.85	76.05	79.76	**92.81**	85.58	73.42	92.09	85.51
10	84.77	77.78	93.43	96.49	**98.52**	78.15	87.15	91.89	96.49	91.48
11	76.69	83.47	51.74	73.36	**98.55**	91.66	87.83	93.73	72.16	87.78
12	72.28	73.50	61.15	75.39	94.35	92.53	71.31	84.83	**97.83**	95.29
13	**93.23**	56.22	45.98	90.66	70.06	62.76	53.74	63.74	78.59	76.95
14	**99.92**	81.80	74.61	94.00	81.92	92.95	94.47	74.24	94.01	92.32
15	**99.76**	77.00	94.81	86.51	91.95	76.43	59.50	92.69	78.66	75.86
16	92.66	99.15	99.97	89.79	95.25	98.47	99.91	**99.98**	98.94	98.87
OA(%)	85.35	86.63	76.16	86.52	89.71	90.43	91.01	92.32	91.67	**93.36**
AA(%)	87.57	76.77	67.95	84.58	87.14	81.72	82.70	83.79	87.64	**87.94**
Kappa	0.8420	0.8443	0.7260	0.8502	0.8804	0.8952	0.8884	0.9105	0.8968	**0.9223**

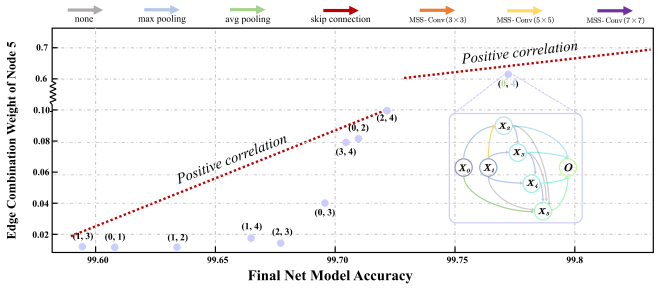


Fig. 10. Optimal structure of cells based on L3M for the KSC dataset, and the topological rating analysis.

accuracy shows a clear positive correlation with the topological weight, highlighting the critical role of accurate topology in enhancing model performance.

C. Comparison of Model Complexity

To address the essential research objectives associated with building lightweight models, we conducted a comprehensive comparison of the trained models against the other methods across the four datasets. This evaluation considered several critical metrics, including the number of model parameters, FLOPs, GPU memory usage, and the time required to infer all of the test set.

As illustrated in Table VIII, the number of parameters and FLOPs serve as direct indicators of the complexity and scale of the model's internal architecture. In addition, GPU memory usage is intricately linked to the batch size, as larger batch sizes during inference necessitate greater GPU memory to achieve higher inference speeds. Consequently, it is evident that the number of parameters and FLOPs does not correlate directly with GPU memory usage.

Upon analyzing Table VIII, we observed that the model derived from the L3M search strategy consistently achieves the highest accuracy across all the datasets while simultaneously maintaining a lower number of parameters and FLOPs. The lightweight operator design and enhanced search strategy of L3M confer significant advantages in complexity relative to LMSS-NAS and LPCN, both of which employ NAS for lightweight network optimization.

To assess the practical applicability of the L3M model, we also compared the inference times of each model on the test sets derived from the four datasets. The inference time of each model reveals that L3M retains a speed advantage over the other NAS methods, although it is slightly outpaced by the manually designed models such as CLOLN. This analysis underscores that the L3M model effectively meets the requirements for a lightweight design while delivering a high performance, enabling efficient model deployment and inference even under constrained computational resources.

D. Hyperparameter Analysis

In the proposed NAS method, the final architecture of each type of search cell is derived through iterative updates based on the current search strategy, culminating in a network composed of multiple search cells. Consequently, both the quantity and type of the search cells play crucial roles in determining the classification performance of the network. To optimize the architecture for HSI classification, we systematically adjusted the two hyperparameters to investigate their impact on network accuracy and to gain insights into effective parameter tuning.

Regarding the types of search cells, NAS incorporates two categories: normal cells and reduction cells. The primary distinction between these cells lies in whether the input feature map undergoes downsampling to reduce its dimensions. The core network structure proposed in this article primarily consists of two interconnected reduction cells. To validate the rationale behind this design, we included these cells in the NAS search process. As illustrated in Table IX, we devised four combinations while maintaining a constant number of search cells, and evaluated the effectiveness of each model based on classification accuracy on the Pavia University dataset and the WHU-Hi-Hanchuan dataset. The results indicate that the network architecture employing two reduction cells yields the best performance. This can be attributed to the fact that, within a two-layer search cell configuration,

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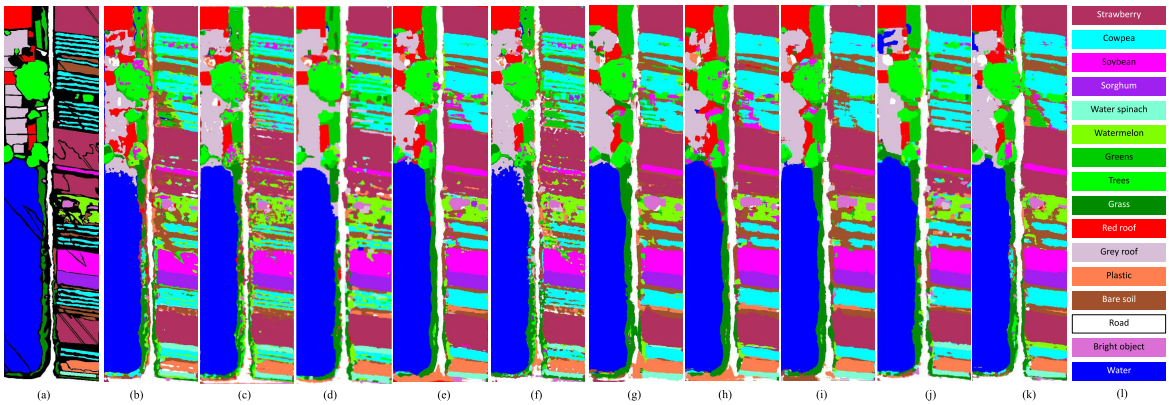


Fig. 11. Classification results of the WHU-Hi-Hanchuan dataset. (a) Ground truth. (b) A^{2}S^{2}K-ResNet. (c) CLOLN. (d) CLOLN-Ghost. (e) LS^{2}CM. (f) LMAFNet. (g) 3D-ANAS. (h) 3D-AT-CNN. (i) LMSS-NAS. (j) LPCN. (k) L3M. (l) Legend.

TABLE VII

EXPERIMENTAL RESULTS ON THE WHU-HI-HONGHU DATASET

Models	A^{2}S^{2}K-ResNet	CLOLN	CLOLN-Ghost	LS^{2}CM	LMAFNet	3D-ANAs*	3D-AT-CNN*	LMSS-NAS*	LPCN*	L3M*
1	99.71	92.58	96.84	89.06	95.90	**99.66**	99.63	99.54	96.55	99.53
2	46.87	84.49	80.60	88.27	**88.94**	46.30	51.05	54.57	66.65	50.91
3	91.00	**97.90**	97.73	94.97	84.50	95.01	97.14	99.82	95.25	96.48
4	98.21	99.52	99.48	99.66	88.12	99.66	99.75	95.54	99.60	**99.69**
5	67.21	52.52	55.43	64.89	**89.25**	82.14	72.73	75.44	83.23	84.75
6	95.39	97.54	91.23	**98.18**	89.54	96.30	97.03	93.14	95.97	97.81
7	88.31	92.94	91.37	93.06	67.20	95.94	92.30	**97.80**	87.57	96.27
8	69.49	35.19	30.05	43.16	63.20	64.55	75.67	73.70	77.45	**83.04**
9	92.53	98.90	96.92	97.83	96.14	98.44	98.10	96.51	**99.24**	97.04
10	96.32	85.21	87.82	81.11	80.84	96.42	91.99	**97.98**	94.07	96.92
11	69.85	74.44	72.27	73.16	79.34	**84.79**	74.93	74.28	82.81	78.17
12	81.42	63.13	52.20	64.23	79.66	76.46	87.25	88.11	**95.27**	90.89
13	84.22	78.85	72.29	81.78	66.78	86.71	89.52	**91.50**	66.44	85.96
14	86.04	74.55	73.49	78.85	78.11	94.09	95.30	93.24	**98.51**	96.99
15	52.23	83.36	62.71	83.22	**97.63**	44.67	48.67	44.22	49.90	51.95
16	96.12	95.73	95.45	**99.84**	95.71	97.37	96.84	98.49	85.92	96.82
17	65.18	93.00	71.27	87.24	**97.38**	76.61	88.76	76.69	87.13	90.45
18	68.39	73.20	**76.68**	72.69	91.43	56.46	72.93	75.25	69.53	68.49
19	83.77	84.44	75.01	90.55	86.62	**90.52**	84.88	83.40	88.74	84.30
20	78.45	72.33	68.96	76.66	88.92	80.13	85.68	**86.97**	85.68	83.71
21	47.98	26.01	20.75	30.02	**95.30**	53.20	63.59	40.84	60.50	58.79
22	52.63	54.96	46.47	57.12	89.10	66.40	67.26	62.84	**90.00**	75.22
OA(%)	88.66	89.42	86.37	90.46	85.21	92.59	93.27	93.32	93.48	**94.10**
AA(%)	76.41	77.76	73.41	79.34	84.74	80.99	83.23	82.46	84.36	**84.89**
Kappa	0.8694	0.8676	0.8804	0.8296	0.8181	0.9068	0.9154	0.9161	0.9210	**0.9257**

TABLE VIII

COMPARISON OF COMPUTATIONAL COMPLEXITY OF DIFFERENT ALGORITHMS ON FOUR DATASETS

Models	A^{2}S^{2}K-RNet	CLOLN	CLOLN-Ghost	LS^{2}CM	LMAFNet	3D-ANAs*	3D-AT-CNN*	LMSS-NAS*	LPCN	L3M*
dataset	PU									
Params/M	0.2217	0.0035	0.0407	0.0106	0.1529	0.2156	0.0537	0.0895	0.0639	**0.0497**
FLOPs/G	0.0878	0.0004	0.0018	0.0004	0.0003	2.0088	0.0198	0.0223	0.0148	**0.0129**
GPU/G	1.718	1.664	1.564	1.542	3.745	1.806	1.749	1.844	1.725	**1.719**
Time/s	13.23	0.21	0.45	0.26	32.75	9.76	11.3	8.47	7.34	**6.35**
dataset	KSC									
Params/M	0.3354	0.0047	0.0521	0.0120	0.1576	0.2243	0.0689	0.0854	0.0691	**0.0613**
FLOPs/G	0.1503	0.0006	0.0027	0.0014	0.0003	3.7915	0.0265	0.0187	0.0134	**0.0122**
GPU/G	1.808	1.556	1.531	1.567	3.114	1.838	1.390	1.156	1.198	**1.209**
Time/s	2.140	0.027	0.203	0.046	9.60	3.173	2.993	2.742	2.467	**1.923**
dataset	WHU-Hi-Hanchuan									
Params/M	0.4197	0.0064	0.0491	0.0203	0.1648	0.2267	0.1091	0.0568	0.0512	**0.0501**
FLOPs/G	0.1953	0.0007	0.0034	0.0013	0.0003	3.0934	0.0290	0.0147	0.0125	**0.0109**
GPU/G	2.037	1.712	1.247	1.227	3.107	1.852	1.815	**1.302**	1.487	1.594
Time/s	106.01	1.74	4.01	2.56	404.90	127.32	83.27	78.29	70.23	**67.15**
dataset	WHU-Hi-Honghu									
Params/M	0.4645	0.0072	0.0562	0.0229	0.1652	0.2567	0.1243	0.0652	0.0602	**0.0576**
FLOPs/G	0.2206	0.0008	0.0039	0.0015	0.0003	3.5266	0.0328	0.0168	0.0158	**0.0125**
GPU/G	2.303	1.941	1.414	1.407	3.202	2.123	2.080	**1.493**	1.646	1.803
Time/s	119.83	1.97	4.53	2.89	772.44	144.02	94.20	88.75	82.13	**76.20**

reduction cells typically involve fewer parameters and less computation, making them more amenable to optimization and training, especially in scenarios characterized by small HSI data volumes or significant data noise. In shallow network

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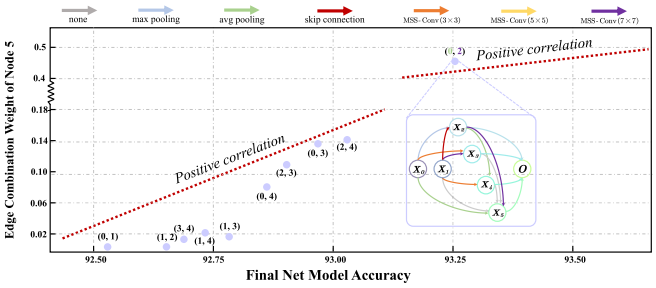


Fig. 12. Optimal structure of cells based on L3M for the WHU-Hi-Hanchuan dataset, and the topological rating analysis.

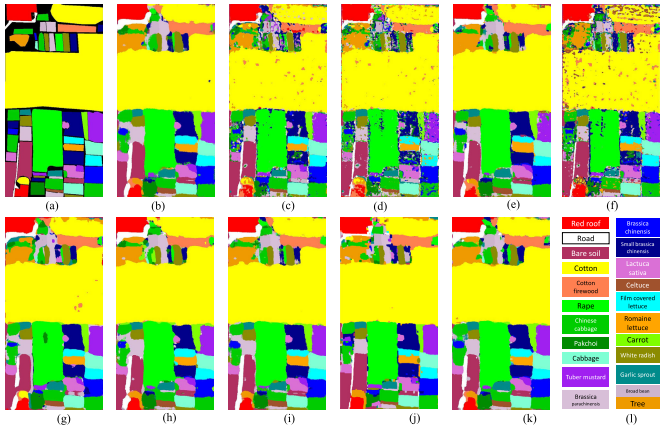


Fig. 13. Classification results of the WHU-Hi-Honghu dataset. (a) Ground truth. (b) A^{2}S^{2}K-ResNet. (c) CLOLN. (d) CLOLN-Ghost. (e) LS^{2}CM. (f) LMAFNet. (g) 3D-ANAS. (h) 3D-AT-CNN. (i) LMSS-NAS. (j) LPCN. (k) L3M. (l) Legend.

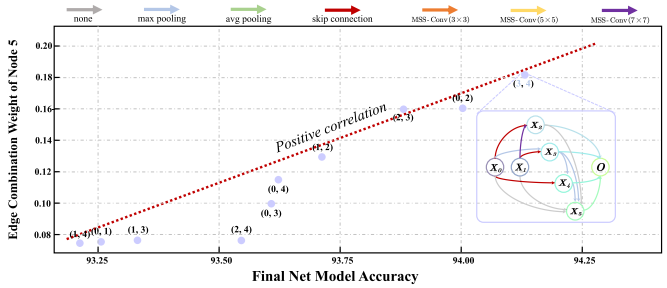


Fig. 14. Optimal structure of cells based on L3M for the WHU-Hi-Honghu dataset, and the topological rating analysis.

designs, it is essential to incrementally enhance the feature abstraction level to capture deeper features, a requirement that is effectively addressed by the use of reduction cells.

Concerning the number of search cells, we conducted comparative experiments with one to four cells. In configurations with one to two search cells, both were designated as reduction cells. For networks with three to four search cells, the second and third cells were assigned as reduction cells, while the remaining cells were classified as normal cells. This strategy was employed to mitigate the information loss in deeper networks and to preserve the intricate patterns and details that can be compromised by excessive downsampling. The experimental results, as listed in Table X, reveal that the OA initially increases with the number of cells before declining, with the optimal performance observed at two cells. This comparative analysis underscores the need to balance network depth with performance in lightweight HSI classification. In particular, the accuracy achieved with a single search cell is comparable to that of networks utilizing three to four search cells, indirectly suggesting that the operators we designed within the candidate operations possess robust feature extraction capabilities.

TABLE IX
ANALYSIS OF HYPERPARAMETERS FOR THE TYPE AND NUMBER OF SEARCHING CELLS

Set.	R-R	R-N	N-N	N-R
dataset	PU			
OA(%)	**98.67**	97.45	97.02	97.28
AA(%)	**97.75**	96.54	96.11	96.37
Kappa	**0.9824**	0.9692	0.9649	0.9675
Set.	R	R-R	$2R$-N	N-$2R$-N
dataset	WHU-Hi-Hanchuan			
OA(%)	93.36	92.12	92.09	91.89
AA(%)	87.94	86.68	86.12	86.50
Kappa	0.9223	0.9103	0.9075	0.9080

E. Ablation Study

1) Efficiency of Operators in the Search Space: We designed a MSS-Conv operator and integrated it into the search space to enhance the feature extraction capability of the candidate operations while maintaining a lightweight model architecture. To evaluate the effectiveness of this operator independently, we substituted it with conventional convolutional layers and several contemporary lightweight modules, including traditional 2-D convolution, inner convolution, GhostNet_Block, COS2M, and Mcs_sepConv. The performance of these alternatives was assessed by comparing the classification accuracy, parameter count, and inference time of the resulting networks.

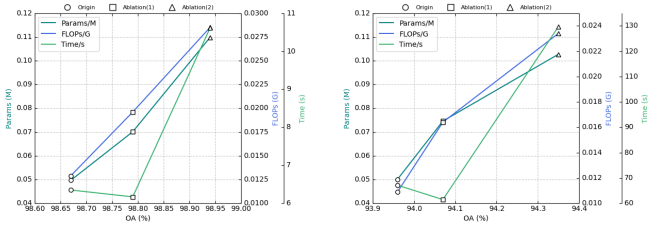
Experiments were conducted on both the Pavia University dataset and the WHU-Hi-Hanchuan dataset, with the findings presented in Table X. The PU dataset was chosen as a representative of traditional hyperspectral datasets, while the Hanchuan dataset represents the more challenging H2 datasets. This selection enables a comprehensive ablation analysis to validate the effectiveness of the designed operators and topological search strategy across different dataset types. The results demonstrate that the proposed MSS-Conv operator not only achieves the highest classification accuracy but also exhibits a significantly lower parameter count and faster inference speed, compared to the alternatives. Compared to the other purpose-built lightweight operators, its inference speed retains a competitive advantage, even when hardware acceleration is not fully utilized. This underscores the effectiveness of the proposed multiscale and multiattention convolution operator in optimizing model performance while adhering to lightweight design principles.

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TABLE X

COMPARISON RESULTS OF DIFFERENT CANDIDATE OPERATIONS UNDER THE SAME SEARCH STRATEGY

dataset	Model	Conv2D	Involution	GhostNet_Block	Mcs_sepConv	COS2M	MSS-Conv
PU	OA(%)	87.92	91.18	95.43	96.29	97.51	**98.67**
	AA(%)	89.88	89.85	94.32	94.26	95.18	**97.75**
	Kappa	0.8462	0.8788	0.9372	0.9510	0.9671	**0.9824**
	Params/M	0.27	0.06	**0.04**	0.09	0.05	0.05
	Time/s	71.56	64.43	8.45	9.8	8.3	**7.71**
WHU-Hi-Hanchuan	OA(%)	82.47	85.19	89.41	92.96	90.44	**93.36**
	AA(%)	79.45	80.07	85.81	85.81	83.27	**87.94**
	Kappa	0.8120	0.8407	0.8633	0.9164	0.8958	**0.9223**
	Params/M	2.2	**0.3**	**0.3**	0.5	0.4	0.4
	Time/s	244.29	80.36	**62.36**	73.10	70.09	67.15



(a) (b)

Fig. 15. Experimental analysis of the impact of lightweight operators on model efficiency and accuracy. (a) Experimental results on Pavia University dataset. (b) Experimental results on WHU-Hi-HanChuan dataset.

2) Analysis of the Impact of Lightweight Design on Model Performance: To further evaluate the impact of each operator on classification accuracy while adhering to the lightweight design principle, we conducted two sets of ablation experiments: 1) removing the equivalent transformation from 3 × 3 convolution to DW-Conv and 2) eliminating the equivalent operation of large receptive field convolution in multiscale stacking. The experiments were performed on the PU dataset and the WHU-Hi-Hanchuan dataset, with the results presented in Fig. 15.

Under the condition that other structural components of the original operator remained unchanged, both ablation operations led to a significant increase in the number of parameters and computational complexity, while the improvement in classification accuracy was relatively limited. Specifically, in ablation setting (1), the standard 3 × 3 convolution replaces the strip convolution. Although this resulted in a substantial increase in the number of parameters and computational complexity, the inference speed was slightly improved due to the extensive optimization of standard convolution in hardware acceleration libraries (e.g., cuDNN). In ablation setting (2), since multiscale feature fusion directly relied on standard convolution operations, inference time increased significantly, further underscoring the importance of lightweight design in computational efficiency. These findings indicate that lightweight operations effectively reduce computational costs while maintaining or even slightly improving classification accuracy, thereby validating the rationality and effectiveness of the proposed method.

3) Ablation Study on Multiscale Feature Extraction: Since the proposed model incorporates the concept of multiscale features, an ablation study was conducted to evaluate the contribution of each scale module to feature extraction. Specifically, the large receptive field operator with feature fusion (7 × 7) was replaced with a single-scale feature operator without feature fusion to assess the independent impact of each scale module on model accuracy. As shown in Table XI, single-scale operators exhibit limited feature extraction capability. When feature fusion is removed, operators with a receptive field similar to the original one experience relatively low accuracy degradation. This is because their required fundamental filter sizes have already been determined upon completion of the architecture search, whereas smaller scale operators primarily supplement fine-grained feature details during fusion. Consequently, it can be concluded that in the feature fusion process, operators at different scales collaborate, where large-scale operators primarily capture global contextual information, while small-scale operators enhance local feature representation, thereby jointly improving the model's representational capacity and classification performance.

TABLE XI

EXPERIMENT ON THE EFFECT OF SINGLE-SCALE FEATURE EXTRACTION

No.	Fusion Scale			OA(%)	
	3 × 3	5 × 5	7 × 7	Pavia University	WHU-Hi-Hanchuan
1	✓			96.12	90.95
2		✓		96.71	91.37
3			✓	97.22	92.63
4	✓	✓	✓	**98.67**	**93.96**

TABLE XII

TOPOLOGICAL SEARCH SUPERIORITY ANALYSIS

Set.	No topology search	With topology search	No topology search	With topology search
Dataset	Pavia University dataset		WHU-Hi-Hanchuan dataset	
OA(%)	98.01	98.67	92.79	93.36
AA(%)	96.81	97.75	81.43	87.94
Kappa	0.9681	0.9824	0.9115	0.9223

4) Effectiveness of Topological Search Strategy: To evaluate the effectiveness of the topology search decoupling strategy, we compared the classification accuracy of the final networks derived from super-network searches conducted in both the first and second stages. Experiments were performed on the Pavia University dataset and the WHU-Hi-Hanchuan dataset, with the results summarized in Table XII. The findings indicate that the classification accuracy of the network structures obtained without implementing the topology search is approximately 1% lower than that of those obtained with the topology search. This underscores the critical importance of establishing the correct topological order during the topology search phase of the L3M framework, thereby validating the necessity of this approach for enhancing network performance.

5) Analysis of the Extraction Capabilities of Attention Mechanisms: Inspired by the class activation mapping (CAM) method, we employ CAM-based heatmaps to analyze the model's sensitivity to specific categories and its feature extraction capability. To evaluate the impact of different attention mechanisms, we conduct an ablation study by

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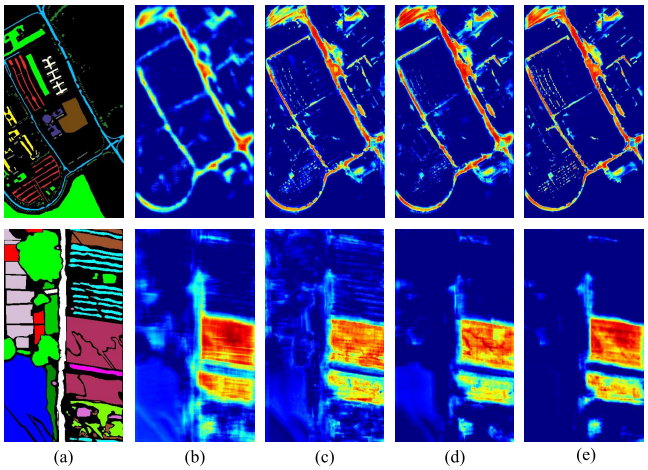


Fig. 16. Visualization of the class activation map. The first row presents heatmaps for class Asphalt in the PU dataset, while the second row corresponds to class Strawberry in the WHU-Hi-Hanchuan dataset. (a) Ground truth labels. (b) Results without channel attention. (c) Results without spatial attention. (d) Results without scale attention. (e) Results from the full model.

individually removing scale attention, spatial attention, and channel attention.

In the PU dataset, class "Asphalt" is characterized by its elongated shape and widespread spatial distribution, whereas in the WHU-Hi-Hanchuan dataset, class "Strawberry" exhibits a more concentrated distribution, with some samples affected by shadow occlusion. As illustrated in Fig. 16, the model incorporating all three attention mechanisms demonstrates a higher focus on the target category, yielding sharper and more distinct heatmap contours. The absence of scale attention reduces the model's ability to capture fine-grained details, while removing spatial attention degrades contour clarity and introduces irrelevant activations. Furthermore, eliminating channel attention significantly weakens the model's ability to focus on the correct category, resulting in blurred heatmaps and a substantial decline in classification accuracy. These findings underscore the critical role of effectively integrating multiple attention mechanisms in enhancing the interpretability and robustness of HSI classification models.

V. Conclusion

In conclusion, the proposed L3M method offers a compelling solution to the challenges faced by the existing DARTS methodologies in HSI classification. By integrating an innovative lightweight classification operator and adopting a multistage search strategy, L3M markedly improves the extraction of multiscale features while effectively addressing the issues of spectral variability and spatial heterogeneity. The deliberate separation of operation search from topology search enhances the accuracy of the topological ratings, thereby reducing the likelihood of suboptimal network structures. Extensive validation across four public datasets confirmed that L3M surpasses the other lightweight models in classification accuracy, all while maintaining a low parameter count, minimized model complexity, and high inference speed. These advancements position L3M as a frontrunner in efficient and effective HSI classification.

Looking ahead, addressing the misclassification of isolated areas often caused by the uneven distribution of training samples will be critical for enhancing the model accuracy and developing a unified search framework for lightweight HSI classification models.

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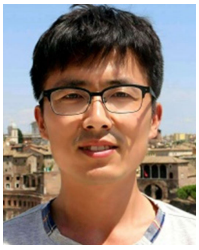
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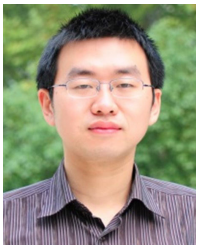
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